

Section B

| Question |     |       |  | Marking details                                                                                                                                                                                                                                                                                                                                                              | Marks available |     |     |       |       |      |
|----------|-----|-------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |  |                                                                                                                                                                                                                                                                                                                                                                              | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 9        | (a) | (i)   |  | 121 kJ mol <sup>-1</sup>                                                                                                                                                                                                                                                                                                                                                     |                 | 1   |     | 1     |       |      |
|          |     | (ii)  |  | $\Delta H = -(-728) - (242) - (1735) - (178) + (-795)$ (1)<br>$\Delta H = -2222$ kJ mol <sup>-1</sup> (1)                                                                                                                                                                                                                                                                    |                 | 2   |     | 2     | 1     |      |
|          |     | (iii) |  | award (1) for any value in range 900-1600 kJ mol <sup>-1</sup><br><br>award (1) for either of following <ul style="list-style-type: none"><li>ionisation energies increase for successive ionisation energies (so value must be more than half the ionisation of the first two electrons)</li><li>second ionisation energy is greater than first ionisation energy</li></ul> |                 | 1   | 1   | 2     |       |      |